1. There are 22 people in a room. 4 are men, 10 are women.

a. What proportion are children? Give your answer as a fraction in its simplest form. _____

b. What proportion are adults? Give your answer as a fraction in its simplest form. _____

If of the 22 people six had red hair and a further 8 were wearing hats:

c. What proportion had neither red hair nor a hat? Give your answer as a fraction in its simplest form. _____

d. What proportion had red hair? Give your answer as a fraction in its simplest form. _____

---

2. Correct the following rounded up to two decimal places:

a. 5.267 = _____

b. 3.312 = _____

c. 8.569 = _____

---

3. In the following questions, circle the number which has the highest value:

a.	6	28	15	32		
b.	5	-2	0	312	1056	1003
c.	6.002	6.08	6.005	6.057	6.0089	
d.	-8	-3	-5	-14		

---

4. Express each of the following decimals as fractions in their simplest form.

a. 0.5 = _____

b. 0.8 = _____

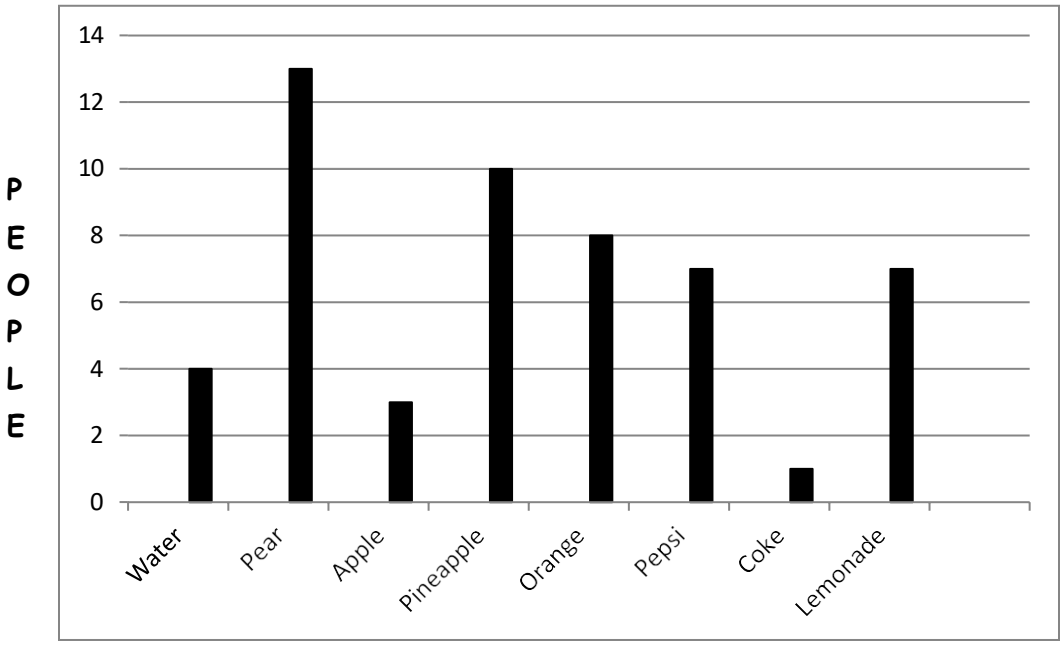
c. 0.065 = _____

d. 0.75 = _____

e. 0.25 = _____

f. 0.79 = _____

5. This chart shows how many people favoured one of 8 different drinks. Study the chart and then answer the questions below.



a. Which type of drink was most popular? _____

b. How many people liked the most popular drink? _____

c. Which drink was least popular? _____

d. How many people liked the least popular type of drink? _____

e. Which two drinks had the same popularity? _____

f. How many people expressed a preference in this study? _____

---

6. Complete the following time questions.

a. Use the 24 Hour clock to write 1.35pm. _____

b. Use the 12 Hour clock to write five to three in the morning. _____

c. Use the 24 Hour clock to write five to three in the morning. _____

d. (I know that my watch is 8 minutes fast.) It shows the time is 10.04pm. What is the correct time using:

1) 12 Hour clock _____

2) 24 Hour clock _____

7. Re-write each of the following correct to two decimal places rounded up:

a.	8.654	_____	**c.**	136.498	_____
b.	16.024	_____	**d.**	542.892	_____

---

8. A prime number can only be divided by itself or 1 exactly. List the numbers below into the correct columns.

3, 5, 6, 9, 11, 19, 27, 36, 43, 51, 54

Prime	Not prime

---

9. Multiply the number shown in each question by 7 and divide the answer by 2.

a.	4 =	_____	**d.**	5 =	_____
b.	8 =	_____	**e.**	12 =	_____
c.	20 =	_____			

---

10. In each question order the numbers by size, then circle the middle number

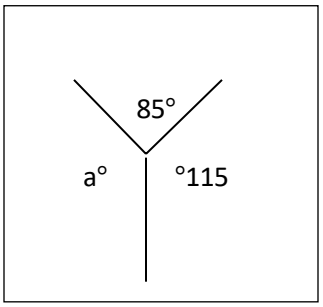
a.	0.228	0.217	0.235	0.227	0.248
b.	-8	+1	+2	+5	-10
c.	9	2	8	50	4
d.	4cm	29mm	209cm	59mm	16cm

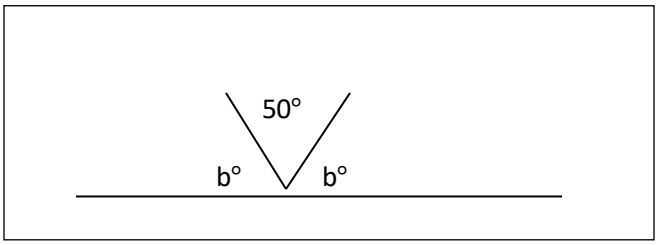
10. (continued)

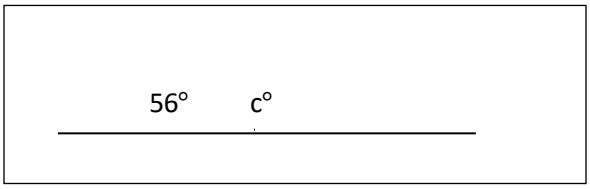
e.	£5.12	£4.82	£3.70	£6.30	£6.32
f.	6g	2g	1.8kg	1.55kg	92g
g.	5^3	2^3	1^2	4^2	3^3
h.	-10	+5	-8	+2	-2
i.	$1\frac{3}{4}$ litres	0.50 litres	0.25 litres	650ml	390ml
j.	$\sqrt{9}$	$\sqrt{79}$	$\sqrt{58}$	$\sqrt{18}$	$\sqrt{102}$

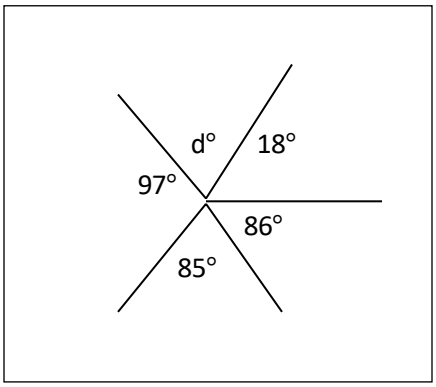
---

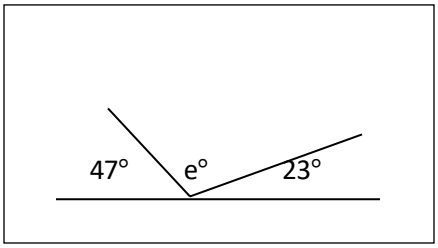
11. Remembering that angles on a straight line add up to 180° and angles round a point add up to 360° calculate the missing angles.











a = _____ **b** = _____ **c** = _____

d = _____ **e** = _____

12. Janice and Cara go out shopping for vegetables together. At the vegetable shop Janice buys 2.63kg of carrots, 3.45kg of potatoes and 1.64kg of tomatoes. Cara buys 2.32kg of potatoes, 2.48kg of parsnips and 3.03kg of tomatoes.

a. Whose shopping was heaviest? _____

b. How much more did Cara's tomatoes weigh? _____

c. What is the total weight of Janice's shopping? _____

d. What weight of potatoes did Janice and Cara buy between them?

_____

e. What weight of tomatoes did Janice and Cara buy between them?

_____

---

13. On a map 1cm represents 15km on the ground. Fill in the gaps below.

		Map	**Actual distance**
	Example:	**2cm**	**30km**
a.		6cm	_____
b.		_____	75km
c.		4cm	_____
d.		_____	150km
e.		12cm	_____
f.		3cm	_____

g. Identify the scale on a new map if 12cm on the map represents 96km.

1 cm to _____ km

_____END OF TEST _____

If you have time

- Go back and complete any questions you missed out.

- Check your answers carefully.

Multiplication to solve problems

The following 2 questions are taken from a Maths paper prepared for 10 year old children. Read through the two following examples of how Long Multiplication can help you solve a problem.

Please make sure of long multiplication first.

DON'T forget to put the nought down when multiplying by 10!

Question one

A computer ran for 35 minutes and 47 seconds.
How many seconds did it run for altogether?

There are 60 seconds in a minute. 35 minutes = 35 x 60

35 x 60 =
2100+47=

Answer 2147 seconds

		3	5	x
		6	0	
		0	0	
2	1	0	0	
2	1	0	0	

Question two

For a party, Mrs George buys 18 packets of crisps that cost 17p each. How much change does she get from £5?

	1	8	x
	1	7	
1	2	6	
1	8	0	
3	0	6	=

£5.00 - £3.06

Answer: = £1.94

You will need to be good at subtraction with money too!

Multiplication with Decimals

- First of all let's revise how to work out a long multiplication sum:

23 x 14 →

I don't know my 14 times table but I *do know* my 10 and 4 times tables.
We can do this sum in 2 stages.

- First we will multiply the units, then the tens.
- Then we add the answers together.

Here is an example:

The yellow highlight shows the 10 of the 14 – cover it with your finger.

```
  23 x
  14
  ----
  92
 230 +
 ----
 322
```

First of all work out the sum 23 x 4 and set your answer out in the usual way.

- If you cover the ten (highlighted in yellow), it will help you to do the sum easily. The answer is 92.
- Now cross out the 4 because you have finished with it.
- Now you must multiply 23 x 10.
- The easiest way is to put down a 0 (highlighted in green) before you start. This is because you are multiplying by 10. Make sure the 0 is under the units column – this is important!
- Simply multiply the other numbers (1 x 23).
- 1 x 3 = 3 1 x 2 = 2

We are now going to take this a stage further by multiplying with decimals.
This sounds very tricky but it is as easy as pie if you know what to do.
The main point to remember is simply to work out your sum as a normal Long Multiplication Sum, <u>ignoring</u> the decimal points.
Pretend that they are not there until you finish the sum.

```
x   2 . 3
    1 . 4
    -----
    9   2
2   3   0 +
-----------
3 . 2   2
```

Now you need to concentrate on those decimal points.
There are 2 orange numbers AFTER the decimal point.
This must be shown in your answer too. The answer becomes **3.22**

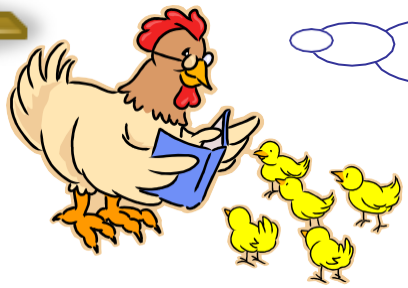
- Now practice these and <u>check your answers on your calculator</u>:

1. 3.4 x 25
2. 5.7 x 52 (only one number here after the decimal point)
3. 3.3 x 2.6 (two numbers after the decimal point)
4. 2.6 x 5.2 (two numbers after the decimal point again)

Equations

Left Hand = Right Hand

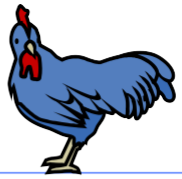
If 9kg are put on one pan, and 4kg and 5kg on the other pan, the scales will balance.



When solving your equations you must **do the same to both sides** of the = sign.
I hope you remember how to work out Positive and Negative Numbers as you will need to be good at them for this work. Do you need to revise them first?

Look at the following equations and follow through the working out carefully. The aim is to find the value of x. Remember that you must do the **same to both sides** to make them balance.

- You must have your helper with you.
- Read through these equations and then you will have some of your own to do.



Handy Hint
When you have found the answer, check by swapping x for that number.
x + 9 = 12
3 + 9 = 12

1. Solve the equation: x + 9 = 12

x + 9 = 12
x = 12 - 9 (Subtract 9 from both sides)
x = 3

2. Solve the equation: 8 + x = 14

8 + x = 14
x = 14 - 8 (Subtract 8 from both sides)
x = 6

3. Solve the equation: x - 2 = 7

x - 2 = 7
x = 7 + 2
x = 9

Fun so far?

4. Solve the equation: x - 15 = 25

$x - 15 = 25$

$x = 25 + 15$ →

$x = 40$

Got it! What I do to one side I must do to the other side.
Here I **add** 15 to the left (making 0) and **add** 15 to the right(making 40).

These next few are trickier Maths, but it becomes fun with lots of practice. I don't think you will get these on your 11+ paper, but they are interesting so have a go! Those of you taking public school entrance had better master it, just in case!

5. Solve the equation: 4x – 3 = 9

Left hand side Right hand side

$4x - 3 = 9$

$4x = 9 + 3$ → Add 3 to both sides

$4x = 12$ → Divide each side by 4

$x = 3$

6. Solve the equation: 2x + 4 = 12

$2x + 4 = 12$

$2x = 12 - 4$ → Subtract 4 to both sides

$2x = 8$ → Divide each side by 2

$x = 4$

7. Solve the equation: 5x - 5 = 25

$5x - 5 = 25$

$5x = 25 + 5$

$5x = 30$

$x = 6$

8. Solve the equation: 2x - 7 = 3

$2x - 7 = 3$

$2x = 3 + 7$

$2x = 10$

$x = 5$

9. Solve the equation: $6x - 8 = 2x$

$$6x - 8 = 2x$$
$$4x - 8 = 0$$
$$4x = 0 + 8$$
$$x = 2$$

10. Solve the equation: $\frac{x}{2} - 5 = 7$

$$\frac{x}{2} - 5 = 7$$
$$\frac{x}{2} = 7 + 5$$
$$\frac{x}{2} = 12$$
$$x = 24$$

Question **10** – Remember:

- Add 5 to both sides
- Then multiply both sides by 2

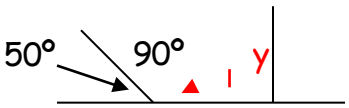
Now have a go on your own! These are easier ones so you should be OK!

1. $3 + x = 7$ **x** = _____
2. $14 - b = 3$ **b** = _____
3. $x + 6 = 18$ **x** = _____
4. $c - 10 = 5$ **c** = _____
5. $5 + y = 12$ **y** = _____
6. $23 - x = 16$ **x** = _____
7. $y + 9 = 24$ **y** = _____
8. $p - 14 = 14$ **p** = _____
9. $8 + a = 15$ **a** = _____
10. $14 - y = 0$ **y** = _____
11. $b + 6 = 24$ **b** = _____
12. $x - 20 = 0$ **x** = _____

1. **4**
2. **11**
3. **12**
4. **15**
5. **7**
6. **7**
7. **15**
8. **28**
9. **7**
10. **14**
11. **18**
12. **20**

Mathematics Revision Test 25

As you only have 10 questions I expect full marks please!

1. 

Find in degrees the measurement of the angle y _____

*See the note below

2. Write 864cm to the nearest m: _____
Rule: Over half way you round up.
Exactly on half you round up.
Under half and you round down.

3. Reduce $\frac{15}{2}$ to Lowest Terms: _____

4. £3.50 x 8 _____

5. 0.2 0.4 0.6 0.8 What are the next 2 numbers? _____
_____

6. 45 ÷ 9 = 40 ÷ _____

7. **34 . 94** Find the difference between the **4** at **x** and the **4** at **y**:
x **y** _____
Have you noticed the decimal point?
Do you remember what ***find the difference*** means?

8. Add 6^2 and 4^2 _____

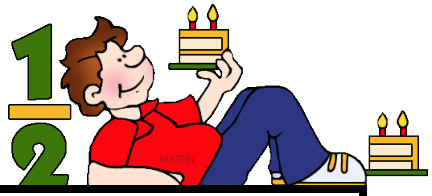
9. Find the average of 5,6,7,8 and 9 _____

10. 0.4 Kg - _____ g = 130 g

*Remember that there are 180° in a straight line.

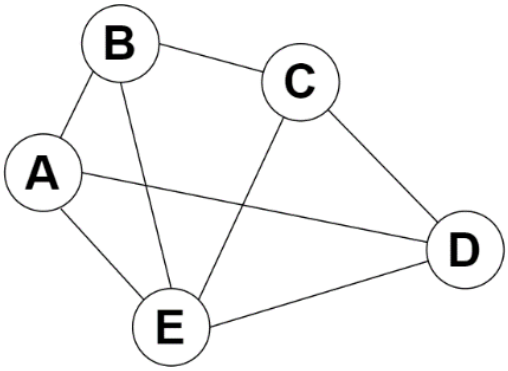
Marks this week: ……

Maths Reasoning



These questions are designed to get your mind whirring - so have a go. If you can't do it first time then try, try again. It is important to persevere.

5 cities are connected by 8 train routes.



The train goes from A through all the other cities before returning back to A.

It visits each of the other cities just once.

How many different possibilities are there?

The costs of each route is given in the table.

A to B	A to D	B to C	B to E	C to D	C to E	D to E	E to A
£4	£5	£8	£6	£5	£6	£7	£5

Which of the routes from A to A is the most expensive?

ANSWERS - COMPLETE MATHS – PART 27

Maths

Maths type 1 paper 2

Page 1

1.
a. 4/11
b. 7/11

c. 4/11
d. 3/11

2.
a. 5.27
b. 3.31
c. 8.57

3.
a. 32
b. 1056
c. 6.08
d. -3

4.
a. 1/2
b. 4/5
c. 13/200
d. 3/4
e. 1/4
f. 79/100

Page 2

5.
a. pear
b. 13
c. coke
d. 1
e. pepsi

and lemonade
f. 53

6.
a. 13:35
b. 2:55am
c. 02:55
d.
1) 9:56pm
2) 21:56

Page 3

7.
a. 8.65
b. 16.02
c. 136.50
d. 542.89

8.
a. prime – 3,5,11,19, and 43
b. not prime - 6,9,27,36, 51 and 54

9.
a. 14
b. 28
c. 70
d. 17.5
e. 42

10.
a. 0.228
b. +1
c. 8
d. 59mm

Page 4

10.
e. £5.12
f. 92gm
g. 4^2
h. -2
i. 0.50litres
j. $\sqrt{58}$

11.
a. 160°
b. 65°
c. 124°
d. 74°
e. 110°

Page 5

12.
a. Cara
b. 1.39kg
c. 7.72kg
d. 5.77kg
e. 4.67kg

13.
a. 90km
b. 5cm
c. 60km
d. 10cm
e. 180km
f. 45km
g. 8km

Long Multiplication with Decimals

1. 85
2. 296.4
3. 8.58
4. 13.52

Equations

(Answers on the sheet)

1. 4
2. 11
3. 12
4. 15
5. 7
6. 7
7. 15
8. 28
9. 7
10. 14
11. 18
12. 20

Revision sheet

1. 40°
2. 9m
3. $\frac{3}{4}$
4. £28
5. 1.0 1.2
6. 8
7. 3.96 (4.00 – 0.04 get it?)
8. 52
9. 7
10. 270g

Maths Reasoning

There are 8 different routes from A to A:

1. A-B-C-D-E-A
2. A-B-C-E-D-A
3. A-B-E-C-D-A
4. A-E-D-C-B-A
5. A-E-B-C-D-A
6. A-D-E-C-B-A
7. A-D-C-E-B-A
8. A-D-C-B-E-A

The costs are:

1.	A-B-C-D-E-A	= 4 + 8 + 5 + 7 + 5 = £29
2.	**A-B-C-E-D-A**	**= 4 + 8 + 6 + 7 + 5 = £30**
3.	A-B-E-C-D-A	= 4 + 6 + 6 + 5 + 5 = £26
4.	A-E-D-C-B-A	= 5 + 7 + 5 + 8 + 4 = £29
5.	A-E-B-C-D-A	= 5 + 6 + 8 + 5 + 5 = £29
6.	**A-D-E-C-B-A**	**= 5 + 7 + 6 + 8 + 4 = £30**
7.	A-D-C-E-B-A	= 5 + 5 + 6 + 6 + 4 = £26
8.	A-D-C-B-E-A	= 5 + 5 + 8 + 6 + 5 = £29

So the routes A-B-C-E-D-A and A-D-E-C-B-A are the most expensive at £30.